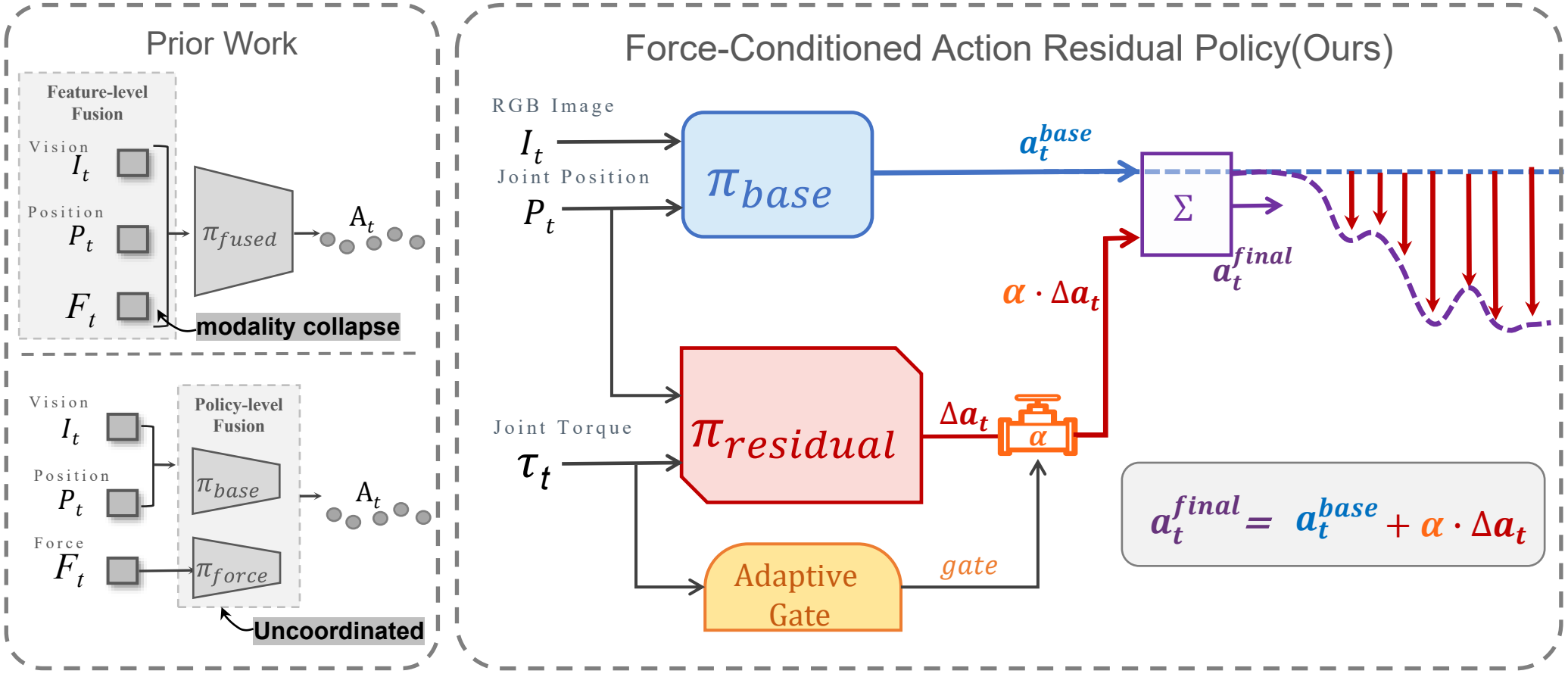
Prior Work
Feature-level Fusion
Vision
I_t
Position
P_t
F_t
π_{fused}
A_t
modality collapse
Vision
I_t
Position
P_t
Force
F_t
Policy-level Fusion
π_{base}
π_{force}
A_t
Uncoordinated
Force-Conditioned Action Residual Policy(Ours)
RGB Image
I_t
Joint Position
P_t
π_{base}
a_t^{base}
Σ
a_t^{final}
$\alpha \cdot \Delta a_t$
Joint Torque
τ_t
$\pi_{residual}$
Δa_t
α
Adaptive Gate
gate
$a_t^{final} = a_t^{base} + \alpha \cdot \Delta a_t$